

Lab Assignment 7

MTH 308: Numerical Analysis and Scientific Computing-I

Department of Mathematics and Statistics, IIT Kanpur

2026 Even Semester: March 27, 2026

This lab will have you implementing the Secant method and Newton's to approximate solutions for several problems. Please write a single script file named `Lab7LastnameRollNumber.m` and publish it as a PDF. Then submit the script `Lab7LastnameRollNumber.m` along with the corresponding published file `Lab7LastnameRollNumber.pdf` (for example, my submitted files would be `Lab7Biswas260102.m` and `Lab7Biswas260102.pdf`) to HELLO IITK.

Each solution should:

- Be contained in a separate cell, including the problem number and a short problem description.
- Be adequately commented to explain your logic.
- Run independently of other cells.

As part of your solution for each problem, output the error tolerance, the approximation, and the number of iterations required, formatted using the `fprintf` function as the sample output below:

Tolerance: 10e-8, Approximation: 1.23456789, Iterations: 23

For a solution accurate to within 10^{-k} , include at least k digits in the output. Unless otherwise noted, use the stopping criteria

$$\left| \frac{x_k - x_{k-1}}{x_k} \right| < \varepsilon.$$

1. Plot the graph of

$$f(x) = x^2 |\sin x| - 4, \quad x \in [0, 4].$$

Use the Secant method to find the smallest positive zero accurate to within 10^{-6} using $p_0 = 3.6$ and $p_1 = 3.7$. What happens if $p_0 = 2.8$ and $p_1 = 2.9$ are used?

2. Use Newton's method to approximate the zero of the function $f(x) = x^2 - 2e^{-x}x + e^{-2x}$ starting from $p_0 = 1$ using accurate to within 10^{-8} . What do you notice about the convergence of Newton's method?
3. Use Newton's method to find a solution to the following nonlinear system. Iterate until

$$\|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\|_\infty < 10^{-6}.$$

$$\begin{aligned} 6x_1 - 2\cos(x_2x_3) - 1 &= 0, \\ 9x_2 + \sqrt{x_1^2 + \sin x_3 + 1.06} + 0.9 &= 0, \\ 60x_3 + 3e^{-x_1x_2} + 10\pi - 3 &= 0. \end{aligned}$$

Use $\mathbf{x}^{(0)} = (0, 0, 0)^t$.